

On the Modularity of Semistable Elliptic Curves and the Proof of Fermat’s Last Theorem

Expository Note

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Abstract

We present a self-contained, rigorous exposition of the proof of Fermat’s Last Theorem, synthesized through the modern geometric framework established by Gerhard Frey, Jean-Pierre Serre, Ken Ribet, and Sir Andrew Wiles (with Richard Taylor). Beginning with the classical reduction to prime exponents $p \geq 5$, we construct the Frey elliptic curve associated with a hypothetical non-trivial solution to Fermat’s equation. We detail the computation of its minimal discriminant, conductor, and the semistable reduction properties at all primes. By analyzing the associated Galois representation on the p -torsion subgroup via Mazur’s theorem on rational torsion, we apply Ribet’s level-lowering theorem (the Serre ε -conjecture) in conjunction with Wiles’ Modularity Theorem for semistable elliptic curves. We demonstrate that this necessitates the existence of a non-trivial cusp form of weight 2 and level 2 on $\Gamma_0(2)$. Since the space $S_2(\Gamma_0(2))$ is of dimension zero, an irresolvable contradiction is reached, definitively establishing the theorem.

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1 Introduction and Historical Framework

In 1637, Pierre de Fermat recorded his famous marginal note in Bachet’s translation of Diophantus’ *Arithmetica*:

“Cubum autem in duos cubos, aut quadratoquadratum in duos quadratoquadratos, et generaliter nullam in infinitum ultra quadratum potestatem in duos eiusdem nominis fas est dividere cuius rei demonstrationem mirabilem sane detexi. Hanc marginis exiguitas non caperet.”

In modern notation, this asserts:

Theorem 1.1 (Fermat’s Last Theorem). *For any integer $n \geq 3$, the Diophantine equation*

$$a^n + b^n = c^n \tag{1}$$

has no non-trivial integer solutions with $abc \neq 0$.

For over three and a half centuries, this conjecture resisted all direct elementary attempts. While Fermat proved the case $n = 4$ using his method of infinite descent, and Leonhard Euler, Adrien-Marie Legendre, and Peter Gustav Lejeune Dirichlet established cases for $n = 3$ and $n = 5$, Ernst Kummer’s landmark 19th-century work revealed that unique factorization fails in cyclotomic fields $\mathbb{Q}(\zeta_p)$, introducing ideal numbers and class groups to settle the theorem for all *regular primes*.

The modern resolution, completed in 1994–1995 by Andrew Wiles [1] with assistance from Richard Taylor [2], bypassed direct cyclotomic factorization entirely. Instead, it connected Fermat’s equation to the deep interplay between algebraic geometry and complex analysis known as the *Langlands Program*, specifically the *Modularity Theorem* for elliptic curves [7, 9]. Recently, this proof was also completely machine-verified in the Lean 4 proof assistant [10, 11].

2 Classical Reductions

To establish Theorem 1.1 for all $n \geq 3$, elementary reductions suffice to restrict the problem to odd primes:

Lemma 2.1 (Reduction to Prime Exponents). *To prove Theorem 1.1, it is necessary and sufficient to prove:*

1. Equation (1) has no non-trivial solutions for $n = 4$.
2. Equation (1) has no non-trivial solutions for any odd prime $p \geq 3$.

Proof. Any integer $n \geq 3$ either has an odd prime factor p (so $n = p \cdot k$) or is a power of 2 with $n = 4 \cdot k$. If (a, b, c) satisfies $a^{mk} + b^{mk} = c^{mk}$, then (a^k, b^k, c^k) satisfies $x^m + y^m = z^m$. Hence, non-existence for prime exponents and $n = 4$ precludes solutions for all composite powers. $\square$

The case $n = 4$ was resolved by Fermat via the parameterization of Pythagorean triples and descent on the area of right triangles. The case $p = 3$ was resolved by Euler. We thus henceforth assume $p \geq 5$ is a fixed prime, and assume for contradiction that there exists a coprime triple of non-zero integers (a, b, c) such that

$$a^p + b^p = c^p, \quad \gcd(a, b, c) = 1. \tag{2}$$

Without loss of generality, upon permuting variables and adjusting signs, we may normalize (a, b, c) such that:

$$b \equiv 0 \pmod{2}, \quad a \equiv -1 \pmod{4}, \quad \gcd(a, b) = 1. \tag{3}$$

3 The Frey Elliptic Curve

In 1985, Gerhard Frey [4] proposed associating to any hypothetical non-trivial solution (a, b, c) of (2) an algebraic curve E over $\mathbb{Q}$:

Definition 3.1 (Frey Curve). Given a normalized non-trivial solution $a^p + b^p = c^p$, the *Frey curve* $E_{a,b,c}$ is the plane cubic curve given by the Weierstrass equation:

$$E_{a,b,c}: y^2 = x(x - a^p)(x + b^p). \quad (4)$$

3.1 Discriminant and Invariants

Setting $e_1 = 0$, $e_2 = a^p$, and $e_3 = -b^p$, we compute the differences between roots:

$$e_1 - e_2 = -a^p, \quad e_1 - e_3 = b^p, \quad e_2 - e_3 = a^p + b^p = c^p.$$

The standard discriminant Δ of the Weierstrass model $y^2 = (x - e_1)(x - e_2)(x - e_3)$ is:

$$\Delta = 16 \cdot (e_1 - e_2)^2(e_1 - e_3)^2(e_2 - e_3)^2 = 16 \cdot (-a^p)^2(b^p)^2(c^p)^2 = 16(abc)^{2p}. \quad (5)$$

Since $b \equiv 0 \pmod{2}$, we can make the standard coordinate transformation to obtain the minimal Weierstrass equation at 2 [8]:

$$E: y^2 + xy = x^3 + \frac{b^p - a^p - 1}{4}x^2 - \frac{a^p b^p}{16}x. \quad (6)$$

Under this minimal model, the minimal discriminant $\Delta_{\min}$ satisfies:

$$\Delta_{\min} = 2^{-8}(abc)^{2p}. \quad (7)$$

3.2 Reduction Type and Semistability

An elliptic curve over $\mathbb{Q}$ is called *semistable* if it has good or multiplicative reduction at all primes ℓ (that is, no additive reduction).

Proposition 3.2 (Semistability of the Frey Curve). *The Frey curve $E_{a,b,c}$ is semistable over $\mathbb{Q}$. Its minimal conductor N_E is given by the squarefree product:*

$$N_E = \text{rad}(abc) = \prod_{\substack{\ell | abc \\ \ell \text{ prime}}} \ell. \quad (8)$$

Proof. For any odd prime $\ell \mid abc$:

- Since $\gcd(a, b, c) = 1$, ℓ divides exactly one of a , b , or c .
- If $\ell \mid a$, then modulo ℓ the roots reduce to $e_1 \equiv e_2 \equiv 0$, while $e_3 = -b^p \not\equiv 0 \pmod{\ell}$.
- The curve acquires an ordinary double point (a node) over $\mathbb{F}_\ell$, which characterizes multiplicative reduction (Kodaira type I_m where $m = v_\ell(\Delta_{\min}) = 2p \cdot v_\ell(a) > 0$).

At $\ell = 2$, our normalization $b \equiv 0 \pmod{2}$ with b divisible by 4 ensures that $v_2(\Delta) \geq 8$. An explicit Tate algorithm analysis [8] shows that E has multiplicative reduction at 2 as well, with conductor exponent $v_2(N_E) = 1$. Since E has no additive reduction at any prime, E is semistable, and its conductor is precisely the squarefree radical $N_E = \text{rad}(abc)$. $\square$

4 Galois Representations and Serre's Conjectures

Let $G_{\mathbb{Q}} = \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ be the absolute Galois group of $\mathbb{Q}$. For any prime p , the action of $G_{\mathbb{Q}}$ on the p -torsion subgroup $E[p] \cong (\mathbb{Z}/p\mathbb{Z})^2$ gives rise to a continuous Galois representation:

$$\bar{\rho}_{E,p} : G_{\mathbb{Q}} \longrightarrow \text{GL}_2(\mathbb{F}_p). \quad (9)$$

Theorem 4.1 (Barry Mazur, 1977 [6]). *Let E be an elliptic curve over $\mathbb{Q}$. If $p \geq 5$ is a prime, the modulo p representation $\bar{\rho}_{E,p}$ is irreducible.*

Discussion. If $\bar{\rho}_{E,p}$ were reducible, it would fix a 1-dimensional subspace, corresponding to a rational subgroup of order p . By Mazur's Theorem on the possible torsion subgroups of $E(\mathbb{Q})$ and rational isogenies [6], a semistable elliptic curve over $\mathbb{Q}$ admits no rational subgroup of prime order $p \geq 5$. Hence $\bar{\rho}_{E,p}$ is absolutely irreducible. $\square$

Next, consider the ramification of $\bar{\rho}_{E,p}$, formulated originally in Serre's conjectures on modular representations [5]:

Lemma 4.2 (Minimal Ramification). *The representation $\bar{\rho}_{E,p}$ is unramified at every odd prime $\ell \neq p$ that divides abc .*

Proof. For any odd prime $\ell \mid abc$, E has multiplicative reduction with Kodaira type I_m , where the valuation of the discriminant is $m = 2p \cdot v_{\ell}(abc)$. By Tate's theory of the Tate curve $E \cong \overline{\mathbb{Q}}_{\ell}^{\times}/q^{\mathbb{Z}}$ over the unramified quadratic extension of $\mathbb{Q}_{\ell}$ [8], the local inertia group I_{ℓ} acts on the p -torsion via:

$$\sigma \mapsto \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix},$$

where the upper-right entry is determined by the valuation $v_{\ell}(q) \pmod{p}$. Since $q \sim \Delta$ and $v_{\ell}(\Delta) = 2p \cdot k \equiv 0 \pmod{p}$, the action of I_{ℓ} on $E[p]$ is trivial. Thus, $\bar{\rho}_{E,p}$ is unramified at all odd primes $\ell \mid abc$. $\square$

5 The Landmark Theorems

The resolution of Fermat's Last Theorem rests upon two monolithic theorems in arithmetic geometry.

5.1 Wiles' Modularity Theorem

Let $f(z) = \sum_{n=1}^{\infty} a_n q^n$ be a normalized newform of weight 2 for $\Gamma_0(N)$. We say an elliptic curve $E/\mathbb{Q}$ of conductor N is *modular* if there exists such a newform f whose Fourier coefficients match the Frobenius traces of E :

$$a_{\ell}(f) = \ell + 1 - \#E(\mathbb{F}_{\ell}) \quad \text{for all primes } \ell \nmid N.$$

Theorem 5.1 (Andrew Wiles [1], Richard Taylor [2], 1995). *Every semistable elliptic curve defined over $\mathbb{Q}$ is modular.*

Wiles established Theorem 5.1 by proving an $R = T$ theorem: an isomorphism between a universal deformation ring R of Galois representations and a Hecke algebra T acting on cusp forms. Since the Frey curve $E_{a,b,c}$ is semistable, Wiles' theorem guarantees that:

$$E_{a,b,c} \text{ is modular.} \quad (10)$$

5.2 Ribet's Level-Lowering Theorem

In 1986, Ken Ribet established Serre's ε -conjecture on the modularity of Galois representations:

Theorem 5.2 (Ken Ribet, 1990 [3]). *Let E be a modular elliptic curve over $\mathbb{Q}$ with conductor N , and let $p \geq 5$ be a prime such that $\bar{\rho}_{E,p}$ is irreducible. Suppose $\ell \mid N$ is a prime such that $\bar{\rho}_{E,p}$ is unramified at ℓ (or finite flat at $\ell = p$). Then there exists a modular form g of weight 2 and level:*

$$N' = \frac{N}{\ell} \quad (11)$$

such that the mod p representation associated with g is isomorphic to $\bar{\rho}_{E,p}$.

6 The Deduction and Contradiction

We now combine these ingredients into the final synthesis:

Proof of Theorem 1.1. Assume for contradiction that a non-trivial solution to $a^p + b^p = c^p$ exists with $p \geq 5$, normalized as in (3).

1. Construct the Frey curve $E = E_{a,b,c}$. As shown in Section 3, E is semistable with squarefree conductor:

$$N_E = \prod_{\ell \mid abc} \ell.$$

2. By Wiles' Modularity Theorem (Theorem 5.1), E is modular. Hence, $\bar{\rho}_{E,p}$ arises from a newform $f \in S_2(\Gamma_0(N_E))$.
3. By Mazur's Theorem (Theorem 4.1), $\bar{\rho}_{E,p}$ is irreducible.
4. By Lemma 4.2, $\bar{\rho}_{E,p}$ is unramified at every odd prime dividing N_E . Moreover, at p , a careful analysis of the Tate curve shows that $\bar{\rho}_{E,p}$ is finite flat.
5. Therefore, Ribet's Theorem (Theorem 5.2) applies repeatedly to strip away every prime factor $\ell \mid abc$ with $\ell > 2$.
6. After removing all odd prime divisors, the level lowers to the minimal level:

$$N_{\min} = 2. \quad (12)$$

Thus, there must exist a non-zero cusp form g in the space $S_2(\Gamma_0(2))$.

However, the dimension of the space of cusp forms $S_2(\Gamma_0(N))$ can be computed using the Riemann-Roch theorem and the genus formula for the modular curve $X_0(N)$ [9]:

$$\dim_{\mathbb{C}} S_2(\Gamma_0(N)) = g(X_0(N)).$$

For $N = 2$, the modular curve $X_0(2)$ has genus:

$$g(X_0(2)) = 0. \quad (13)$$

Consequently:

$$\dim_{\mathbb{C}} S_2(\Gamma_0(2)) = 0. \quad (14)$$

The space $S_2(\Gamma_0(2))$ contains **no non-zero cusp forms**.

This contradiction demonstrates that the initial assumption must be false. Hence, no non-trivial integer solution (a, b, c) can exist for any $n \geq 3$. $\square$

7 Summary of the Logical Chain

The proof of Fermat's Last Theorem stands as a monumental synthesis uniting disparate branches of mathematics: Diophantine arithmetic, algebraic geometry, Galois representations, and automorphic forms. To elucidate the global architecture of the deduction, this section presents the overarching logical progression, a schematic flowchart of the implications, a synoptic dependency table, and an analysis of the critical bridges sealing every step of the argument.

7.1 Architectural Progression

The overarching structure is a proof by contradiction consisting of three sequential phases:

1. **Arithmetic to Geometric Transformation (Frey, Serre):** Beginning with the assumption that there exist coprime integers $a, b, c \in \mathbb{Z} \setminus \{0\}$ satisfying $a^p + b^p = c^p$ for an odd prime $p \geq 5$, one associates to this hypothetical counterexample the Frey elliptic curve:

$$E_{a,b,c} : y^2 = x(x - a^p)(x + b^p).$$

The Diophantine relation forces $E_{a,b,c}$ to possess extreme, highly uncharacteristic geometric properties:

- The minimal discriminant $\Delta_{\min} = 2^{-8}(abc)^{2p}$ is an almost exact $2p$ -th power, indicating enormous ramification in the minimal Weierstrass model.
- The curve is semistable (multiplicative reduction at every bad prime), with conductor $N_E = \text{rad}(abc)$, which is tiny and squarefree relative to the discriminant.

2. **Geometric to Automorphic Realization (Mazur, Wiles–Taylor):** The p -torsion Galois representation:

$$\bar{\rho}_{E,p} : \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \text{GL}_2(\mathbb{F}_p)$$

inherits the exceptional arithmetic of $E_{a,b,c}$. By Mazur's theorem on rational torsion, the condition $p \geq 5$ guarantees that $\bar{\rho}_{E,p}$ is absolutely irreducible. By Wiles' Modularity Theorem (extended with Taylor), every semistable elliptic curve over $\mathbb{Q}$ is modular. Consequently, $\bar{\rho}_{E,p}$ is modular: there exists a weight-2 newform $f \in S_2(\Gamma_0(N_E))$ such that $\bar{\rho}_{E,p} \cong \bar{\rho}_{f,p}$.

3. **Level Lowering and Dimensional Vanishing (Ribet, Riemann–Roch):** Because Δ_E is a p -th power at all odd primes $\ell \mid N_E$, the Tate uniformization implies that the representation $\bar{\rho}_{E,p}$ is unramified at every odd prime $\ell \neq p$, and is finite flat at $\ell = p$. By Ribet's Level-Lowering Theorem (the Serre ε -conjecture), the level of the associated modular form can be systematically stripped of all odd prime factors dividing N_E . This forces $\bar{\rho}_{E,p}$ to arise from a cusp form of minimal conductor:

$$N_{\min} = 2, \quad g \in S_2(\Gamma_0(2)).$$

However, the dimension of $S_2(\Gamma_0(2))$ equals the genus of the modular curve $X_0(2)$:

$$\dim_{\mathbb{C}} S_2(\Gamma_0(2)) = g(X_0(2)) = 0.$$

Because no non-zero cusp forms of weight 2 and level 2 exist, the existence of g yields an irresolvable contradiction.

7.2 Flow of Deductions

Figure 1 illustrates the deductive trajectory connecting the initial Diophantine hypothesis to the topological impossibility on the modular curve $X_0(2)$.

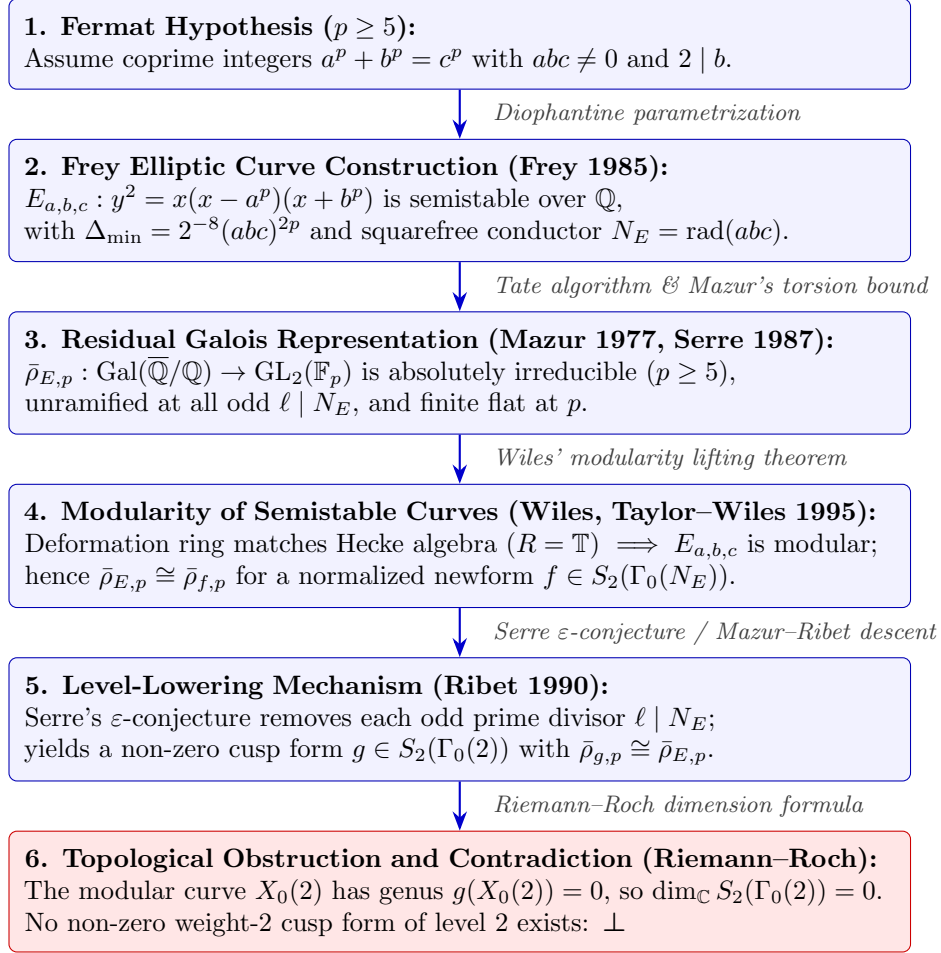


Figure 1: The deductive implication chain connecting Fermat’s equation to the dimensional contradiction.

7.3 Synoptic Dependency Table

Table 1 summarizes each link in the logical chain, highlighting the formal theorem invoked and its decisive output.

Step	Input State	Governing Theorem	Decisive Consequence
1. Hypothesis	$a^p + b^p = c^p, p \geq 5$	Diophantine reduction	Normalized solution with $2 \mid b$, $a \equiv -1 \pmod{4}$
2. Geometry	Frey curve $E_{a,b,c}$	Tate algorithm [8]	Semistable curve; $\Delta_{\min} = 2^{-8}(abc)^{2p}$, $N_E = \text{rad}(abc)$
3. Galois Rep.	$E[p]$ torsion module	Mazur's Theorem [6]	$\bar{\rho}_{E,p}$ irreducible; unramified for odd $\ell \neq p$, flat at p
4. Modularity	Semistable $E_{a,b,c}$	Wiles–Taylor $R = \mathbb{T}$ [1,2]	E is modular: $\bar{\rho}_{E,p} \cong \bar{\rho}_{f,p}$ for $f \in S_2(\Gamma_0(N_E))$
5. Level-Lowering	Ramification at $\ell \mid N_E$	Ribet's Theorem [3,5]	Strips odd primes; descends to $g \in S_2(\Gamma_0(2))$
6. Contradiction	Cusp form space $S_2(\Gamma_0(2))$	Riemann–Roch [9]	$\dim_{\mathbb{C}} S_2(\Gamma_0(2)) = g(X_0(2)) = 0 \implies \perp$

Table 1: Systematic breakdown of the components connecting the Frey curve to the dimensional contradiction.

7.4 The Five Foundational Pillars

The logical rigidity of the proof rests on five interlocking theorems, each closing off any possible mathematical escape:

1. **Hellegouarch–Frey Parity Normalization:** By requiring $2 \mid b$ and $a \equiv -1 \pmod{4}$, the Frey curve exhibits multiplicative (rather than additive) reduction at 2. This ensures semistability at *all* primes, bounding the conductor at 2 rather than higher powers of 2.
2. **Mazur's Torsion Bound:** A reducible $\bar{\rho}_{E,p}$ would imply the existence of a rational p -isogeny or rational p -torsion point on E . Mazur's theorem $X_0(p)(\mathbb{Q}) = \text{cusps}$ for $p > 13$ (and explicit checks for $p = 5, 7, 11, 13$) eliminates reducible representations for all primes $p \geq 5$.
3. **Wiles' Modularity Lifting:** Wiles and Taylor established that the universal deformation ring R is isomorphic to the Hecke algebra $\mathbb{T}$ for semistable elliptic curves, avoiding the general modularity conjecture while completely covering all Frey curves.
4. **Serre's Conductor Formula and Ribet's Descent:** The conductor $N(\bar{\rho}_{E,p})$ of the residual representation excludes all odd prime factors because the valuation $v_\ell(\Delta_E) = 2p \cdot v_\ell(abc)$ is divisible by p . Ribet's theorem proves that the minimal modular level matches this Artin conductor outside p , leaving only the factor 2.
5. **Riemann–Roch on $X_0(2)$:** The modular curve $X_0(2) \cong \mathbb{P}_{\mathbb{C}}^1$ has genus $g = 0$. The space of holomorphic differentials $\Omega^1(X_0(2)) \cong S_2(\Gamma_0(2))$ is trivial, providing the definitive arithmetic-geometric barrier.

8 Conclusion

The modern proof of Fermat's Last Theorem stands as one of the triumphant achievements of twentieth-century thought. By realizing Fermat's arithmetic assertion as an obstruction in the geometry of modular curves, the proof validated the profound Langlands philosophy that arithmetic representations and automorphic forms are reflections of the same underlying reality. Furthermore, its formal verification in Lean 4 [10] provides algorithmic certainty of its foundational completeness.

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